

Notes: Bacchetta, van Wincoop & Young (RES, 2023)
Infrequent Random Portfolio Decisions in an Open Economy Model

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1 Big picture

- **Question.** How can an open-economy DSGE model reconcile international equity portfolios, asset-price movements, excess-return predictability, and momentum/reversal when frictionless portfolio models imply unrealistically fast portfolio adjustment?
- **Core idea.** Investors do not continuously re-optimize portfolios. Each period, only a random fraction p of investors makes a new portfolio decision, analogous to Calvo price setting. The remaining investors keep portfolio shares fixed.
- **Key mechanism.** Infrequent decision-making makes aggregate portfolios persistent and makes current portfolio choices forward-looking over many future excess returns rather than only the next-period excess return.
- **Empirical target.** The paper applies the model to monthly equity portfolios and returns for the U.S. versus the rest of the world, with additional panel evidence across 73 countries.
- **Main result.** An intermediate adjustment probability, especially $p = 0.1$, fits the broad set of portfolio and return moments best.

2 Incremental contribution

- **Calvo portfolio adjustment.** The paper is the first in this literature to use random Calvo-style portfolio revision in a two-country DSGE portfolio model.
- **Micro-to-macro bridge.** It connects household and fund-level portfolio inertia to macro-finance moments for global equity portfolios and prices.
- **Unified explanation.** The same friction helps explain portfolio inertia, limited sensitivity to expected returns, high price impact of financial flows, return predictability, and momentum/reversal.
- **Novel portfolio expression.** With $p < 1$, portfolio demand depends on lagged portfolios and the present discounted value of all future expected excess returns.
- **Global solution plus approximation.** The full DSGE model is solved globally, while a simple approximation gives transparent intuition.

3 Model objects, data, and measurement

3.1 Two-country environment and assets

There are two countries, Home and Foreign, and a single good. Agents can hold Home equity, Foreign equity, and a one-period risk-free bond.

$$R_{H,t+1} = \frac{Q_{H,t+1} + D_{H,t+1}}{Q_{H,t}}, \quad R_{F,t+1} = \frac{Q_{F,t+1} + D_{F,t+1}}{Q_{F,t}}.$$

Detailed interpretation: The relevant asset-pricing object is the relative return of Home equity versus Foreign equity.

3.2 Dividend and financial shocks

$$\begin{aligned} d_t^D &= \rho_d d_{t-1}^D + \varepsilon_t^{d,D}, & d_t^A &= (1 - \rho_d)d + \rho_d d_{t-1}^A + \varepsilon_t^{d,A}. \\ \tau_t^D &= \rho_\tau \tau_{t-1}^D + \varepsilon_t^{\tau,D}, & \tau_t^A &= (1 - \rho_\tau)\tau + \rho_\tau \tau_{t-1}^A + \varepsilon_t^{\tau,A}. \end{aligned}$$

Detailed interpretation: Financial shocks are reduced-form exogenous portfolio shifts, interpretable as liquidity trade, noise trade, hedge demand, or changes in risk-bearing capacity.

3.3 Data

The monthly sample runs from November 1995 to December 2018. The main empirical comparison is the U.S. versus the rest of the world, an aggregate of 44 countries. MSCI data provide equity prices, total returns, and earnings-based dividends. Portfolio data use U.S. external equity assets and liabilities combined with market capitalization. Predictability regressions use a 73-country panel.

4 Models and theoretical specifications

4.1 Portfolio Euler equations

With infrequent portfolio decisions, the Euler equations contain standard expected-return terms and additional terms that reflect the value of being locked into a portfolio:

$$\begin{aligned} E_t m_{t+1} (e^{\tau_{H,t}} R_{H,t+1} - R_t) + (1-p) E_t m_{t+1}^o R_{t+1}^p \lambda_{HH,t+1} &= 0, \\ E_t m_{t+1} (e^{-\tau_{H,t}} R_{F,t+1} - R_t) + (1-p) E_t m_{t+1}^o R_{t+1}^p \lambda_{HF,t+1} &= 0. \end{aligned}$$

Detailed interpretation: When $p = 1$, the second terms disappear. When $p < 1$, today's decision accounts for future periods when the investor may not be able to revise.

4.2 Approximate aggregate portfolio rule

$$\begin{aligned} z_t^{e,A} &= (1-p) z_{t-1}^{e,A} + \frac{p}{(1-\theta)D} \sum_{i=1}^{\infty} (1-\theta) \theta^{i-1} E_t e r_{t+i} + 0.5 f_t, \\ f_t &= \frac{2p}{(1-\theta)D} \tau_t^D. \end{aligned}$$

Detailed interpretation: Current aggregate portfolio demand is the sum of lagged portfolios, a present value of future excess returns, and an exogenous financial-flow shock.

4.3 Price impact and predictability

$$\begin{aligned} M &= \frac{q_t^D}{f_t}. \\ e r_{t+1,t+i} &= a_1 (d_t^D - q_t^D) + u_t, \quad q_{t+3}^D - q_t^D = a_1 (q_t^D - q_{t-1}^D) + a_2 q_t^D + u_t. \end{aligned}$$

Detailed interpretation: The first equation measures dividend-yield predictability. The second summarizes momentum and reversal.

5 Calibration, validation, and theoretical design

The authors compare $p = 0.04, 0.1, 0.2, 1$ and a $p = 1$ small-shock case. Financial-shock parameters are recalibrated for each p to target key moments. The main design is over-identification by many moments: a successful friction must jointly match portfolio persistence, price impact, excess-return volatility, predictability, and momentum/reversal.

6 Section-by-section study guide

- Section 2 builds the two-country DSGE environment with random portfolio decision opportunities.
- Section 3 derives the approximate portfolio expression.
- Section 4 describes the U.S.-ROW data and calibration.
- Section 5 compares model moments for different values of p .
- Appendices validate the approximation and the stationarity of relative wealth.

7 Tables and figures

7.1 Table 1: Calibrated parameters

Read: The table reports five parameterizations. Dividend parameters are common across columns, while financial-shock parameters vary to target key moments for each p .

Detailed interpretation: The table is the model's calibration design. It gives every p value a fair chance by recalibrating financial shocks.

Economic message: The comparison across columns isolates the economic role of the portfolio friction.

TABLE 1
Calibrated parameters

Parameters	$p=0.04$	$p=0.1$	$p=0.2$	$p=1$	$p=1$ smaller financial shock
γ	10	10	10	10	10
β	0.99668	0.99668	0.99668	0.99668	0.99668
ρ_d	0.9767	0.9767	0.9767	0.9767	0.9767
σ_{dp}	0.0447	0.0447	0.0447	0.0447	0.0447
σ_{dA}	0.0325	0.0325	0.0325	0.0325	0.0325
ρ_r	0.98084	0.99003	0.99151	0.99177	0.99177
τ	0.000941	0.0001050	0.0001011	0.0001078	0.0000410
$\sigma_{r,D}$	0.0006514	0.0003851	0.0003387	0.0003442	0.0000020
$\sigma_{r,A}$	0.0000624	0.0000359	0.0000243	0.0000084	0.0000084

Notes: The table reports parameters under five parameterizations. The first four parameterizations set the parameters of the financial shock process (the last four parameters) to target four moments. The other parameters remain the same. The last column of parameters is the same as the previous column, except that $\sigma_{r,D}$ is lowered such that the standard deviation of a financial shock innovation $\Delta f_t = \frac{2p}{(1-p)D} \epsilon_t^{r,D}$ is the same as under the $p=0.1$ parameterization and τ is adjusted to match the observed portfolio home bias.

Table 1: Calibrated parameters

TABLE 2
Portfolio terms and price impact

	$p=0.04$	$p=0.1$	$p=0.2$	$p=1$	$p=1$ smaller financial shock
Features portfolio expression					
Coefficient lagged portfolio	0.96	0.9	0.8	0	0
Coefficient expected excess return	5.6	15.1	30.7	147.1	2,943
Financial shock					
s.d. financial shock innovation Δf	0.0073	0.0117	0.0208	0.1012	0.0117
Price impact M	3.44	2.15	1.20	0.25	0.0146
Fraction $var(er_{t+1})$ due to	0.977	0.956	0.944	0.924	0.00
Financial shock					

Notes: The table reports information about the three elements of the optimal portfolio expression $z_t^{r,A} = (1-p)z_{t-1}^{r,A} + \frac{p}{(1-p)D} \sum_{i=1}^{\infty} (1-\theta)^{i-1} E_t er_{t+i} + 0.5f_t$. The coefficient on the lagged portfolio is $1-p$. The coefficient on the weighted average of expected future excess returns (with weights adding to 1) is $\frac{p}{(1-p)D}$. For the last portfolio term, the table reports the standard deviation of the financial shock innovation $\Delta f_t = \frac{2p}{(1-p)D} \epsilon_t^{r,D}$, which is the size of the exogenous financial flow between the two equity as a fraction of either equity market. It also reports the price impact M , which is the ratio of the relative price change to the change in Δf_t in response to a financial shock innovation $\epsilon_t^{r,D}$. The last row reports the fraction of the variance of the excess return that is associated with financial shocks. The columns correspond to the parameterizations reported in Table 1.

Table 2: Portfolio terms and price impact

7.2 Table 2: Portfolio terms and price impact

Read: The lagged portfolio coefficient is $1-p$; the expected-return coefficient explodes in the frictionless case; price impact is much larger with infrequent adjustment.

Detailed interpretation: With $p < 1$, portfolios are sticky and financial flows move prices. With $p = 1$, investors absorb flows too elastically, so prices move too little.

Economic message: Infrequent portfolio choice makes financial flows price-relevant.

7.3 Table 3: Data and model moments

Read: The $p = 0.1$ model matches excess-return volatility, portfolio-share volatility, autocorrelations, and correlations better than the very sticky or frictionless alternatives.

Detailed interpretation: This table is the central validation table. It shows that an intermediate friction fits the broad moment set.

Economic message: The data reject both nearly frozen portfolios and fully frictionless portfolios.

TABLE 3
Data and model moments

	Data		Model			
	$p=0.04$	$p=0.1$	$p=0.2$	$p=1$	$p=1$ smaller financial shock	
Standard deviations						
er_t	0.0262	0.0257 (0.0014)	0.0258 (0.0010)	0.0260 (0.0010)	0.0262 (0.0011)	0.0058 (0.0003)
$z_t^{e,A}$	0.0297	0.0322 (0.0090)	0.0266 (0.0080)	0.0247 (0.0080)	0.0253 (0.0074)	0.0052 (0.0019)
$\Delta z_t^{e,A}$	0.0046	0.0054 (0.0004)	0.0052 (0.0004)	0.0050 (0.0003)	0.0051 (0.0003)	0.0015 (0.0003)
$\Delta z_t^{e,D}$	0.0048	0.0047 (0.0008)	0.0046 (0.0006)	0.0046 (0.0004)	0.0046 (0.0002)	0.0974 (0.0041)
Autocorrelations						
er_t	0.0686	0.3765 (0.0559)	0.1022 (0.0605)	0.0294 (0.0590)	0.0181 (0.0601)	-0.0039 (0.0592)
$z_t^{e,A}$	0.9807	0.9828 (0.0107)	0.9741 (0.0162)	0.9710 (0.0190)	0.9721 (0.0182)	0.9665 (0.0224)
$\Delta z_t^{e,A}$	0.0971	0.3784 (0.0604)	0.0968 (0.0729)	0.0170 (0.0690)	0.0199 (0.0608)	-0.0072 (0.0692)
q_t^D	0.9803	0.9826 (0.0103)	0.9751 (0.0137)	0.9720 (0.0170)	0.9729 (0.0175)	0.9593 (0.0199)
Contemporaneous correlations						
$\text{corr}(\Delta z_t^D, er_t)$	0.2025	0.1155 (0.0602)	0.2044 (0.0572)	0.2304 (0.0577)	0.2351 (0.06)	0.9963 (0.0017)
$\text{corr}(\Delta z_t^D, \Delta z_t^{e,A})$	0.2709	0.1361 (0.0586)	0.2001 (0.0572)	0.2286 (0.0578)	0.2361 (0.0602)	0.9375 (0.0854)

Notes: Model moments and associated standard errors (in parentheses) are based on 1,000 simulations of a 278 month period. Results are shown for the parameterizations reported in Table 1. In the last column, the size of a one standard deviation of the financial shock innovation Δf_t is the same as under $p=0.1$. Of the four targeted moments, the table only reports the standard deviations of the excess return and $\Delta z_t^{e,D}$ which are indicated in *italics*.

Table 3: Data and model moments

7.4 Table 4: Excess return predictability

Read: The $p = 0.1$ model matches dividend-yield predictability and the momentum/reversal regression coefficients closely.

Detailed interpretation: Slow portfolio adjustment creates humped price responses, which generate short-run momentum and longer-run reversal.

Economic message: The same friction explains both portfolio inertia and return predictability.

7.5 Figure 1: Impulse response functions

Read: Figure 1 shows responses of relative price and average portfolio shares to dividend and financial shocks. Low and intermediate p generate delayed overshooting.

TABLE 4
Excess return predictability

	Data		Model			
	$p=0.04$	$p=0.1$	$p=0.2$	$p=1$	$p=1$ smaller financial shock	
Predictability by relative dividend yield						
$er_{t+1} = a_1(d_t^D - q_t^D)$	0.0066 (0.0020)	0.0078	0.0068	0.0066	0.0068	0.0000
$er_{t+1,t+3} = a_1(d_t^D - q_t^D)$	0.0185 (0.0038)	0.0297	0.0216	0.0201	0.0205	0.0000
$er_{t+1,t+12} = a_1(d_t^D - q_t^D)$	0.0779 (0.0090)	0.1283	0.0849	0.0775	0.0764	0.0000
Momentum and reversal						
$q_{t+3}^D - q_t^D = a_1(q_t^D - q_{t-1}^D) + a_2 q_t^D$	0.1020 (0.0296)	0.5740	0.1158	0.0350	0.0046	-0.0034
a_1	0.1020 (0.0296)	0.5740	0.1158	0.0350	0.0046	-0.0034
a_2	-0.0290 (0.0045)	-0.0520	-0.0287	-0.0260	-0.0260	-0.0513
Expected excess return						
s.d. theoretical $E(er)$	0.0106	0.0035	0.0035	0.0024	0.0024	0.0000
a.c. theoretical $E(er)$	0.3980	0.5311	0.5311	0.9527	0.9917	0.9985
s.d. regression-based $E(er)$	0.0042 (0.0011)	0.0105	0.0035	0.0024	0.0026	0.0000
a.c. regression-based $E(er)$	0.4814 (0.2020)	0.4335	0.5308	0.9470	0.9824	0.9223

Notes: Data moments are based on panel regressions for 73 countries, with standard errors in parenthesis. Model moments are population moments computed by simulating the model over 200,000 years. Results are shown for the

Table 4: Excess return predictability

Detailed interpretation: The humped price response is the visual source of momentum and reversal.

Economic message: Slow portfolio adjustment converts shocks into delayed price movements.

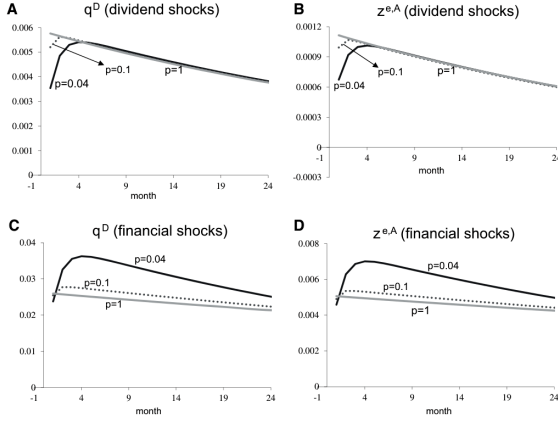


FIGURE 1
Impulse response functions
Notes: The charts show the impulse response functions of q^D and $z^{e,A}$ in response to a one standard deviation increase in $\varepsilon^{d,D}$ (dividend shock) and $\varepsilon^{f,D}$ (financial shock), starting from the symmetric steady state.

Figure 1: Impulse response functions

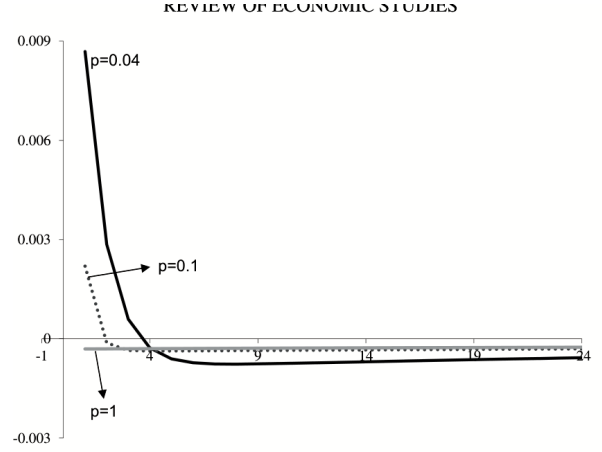


FIGURE 2
Impulse response expected excess return (financial shocks)
d excess returns are computed from impulse response functions. Excess returns in the periods subsequent to the

Figure 2: Impulse response expected excess return

7.6 Figure 2: Impulse response expected excess return

Read: Figure 2 shows expected excess returns after a financial shock. For $p = 0.04$ and $p = 0.1$, expected returns are initially positive and later negative.

Detailed interpretation: The sign change is the return-side version of delayed overshooting.

Economic message: The model endogenously generates momentum followed by reversal.

7.7 Appendix Figures A1 and A2

Read: Figure A1 validates the approximate portfolio rule; Figure A2 shows a stationary ergodic distribution for relative wealth.

Detailed interpretation: The approximation is close to the global solution, and the model has stable long-run dynamics.

Economic message: The transparent portfolio rule is a reliable guide to the numerical model.

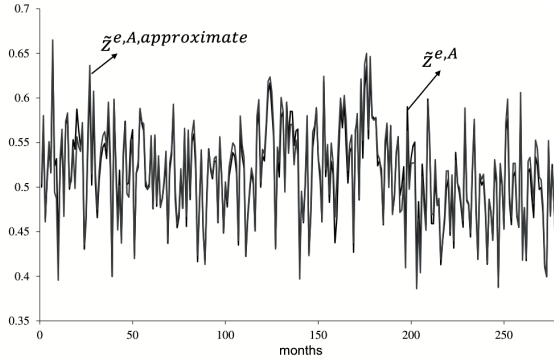


FIGURE A1
Approximation optimal portfolio

Notes: The chart shows the global solution of the optimal portfolio based on a simulation over 278 months, as well as the approximation (34), without the hedge term h_t . The correlation between the global solution and the approximation is 0.976.

(a) Figure A1

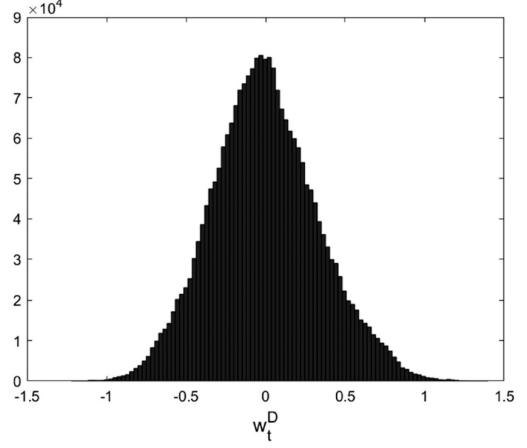


FIGURE A2
Ergodic distribution relative wealth

(b) Figure A2

8 Result map

- Portfolio inertia comes from the lagged portfolio term $1 - p$.
- Limited sensitivity to expected returns comes from a present-value portfolio rule rather than one-period reoptimization.
- Price impact rises because only a subset of investors adjusts immediately.
- Momentum and reversal arise from humped price responses.
- $p = 0.1$ is the best-fitting intermediate friction.
- The frictionless model fails on expected-return sensitivity, price impact, and momentum/reversal.

9 Short summary

- The paper embeds random infrequent portfolio decisions in a two-country DSGE model.
- The model is calibrated to U.S. versus ROW monthly equity portfolios and returns from 1995 to 2018.
- The preferred $p = 0.1$ implies portfolio revision roughly once every ten months.
- The model matches portfolio inertia, excess-return volatility, price impact, predictability, momentum, and reversal.
- The frictionless model reacts too strongly to expected returns and generates too little price impact.

10 Conclusion

10.1 Core conclusion

Infrequent random portfolio decisions are a quantitatively important friction for open-economy asset pricing. Slow-moving portfolios help explain both portfolio dynamics and asset-price dynamics.

10.2 Economic intuition

Only a fraction of investors adjusts immediately after a shock. Prices must move more to induce active investors to absorb flows. As more investors later adjust, portfolios catch up and prices partially reverse.

10.3 Incremental contribution

The paper turns portfolio inertia into a general-equilibrium asset-pricing mechanism and shows that a Calvo-style portfolio friction can unify portfolio behavior, price impact, and excess-return predictability.

10.4 Limitations

Financial shocks are reduced-form, the model aggregates the rest of the world, and the adjustment probability is constant. Future work could connect the financial shocks to institutions or investor types.

10.5 Takeaway

Portfolio inertia is not a minor detail; it changes the equilibrium transmission of capital flows into prices and returns.